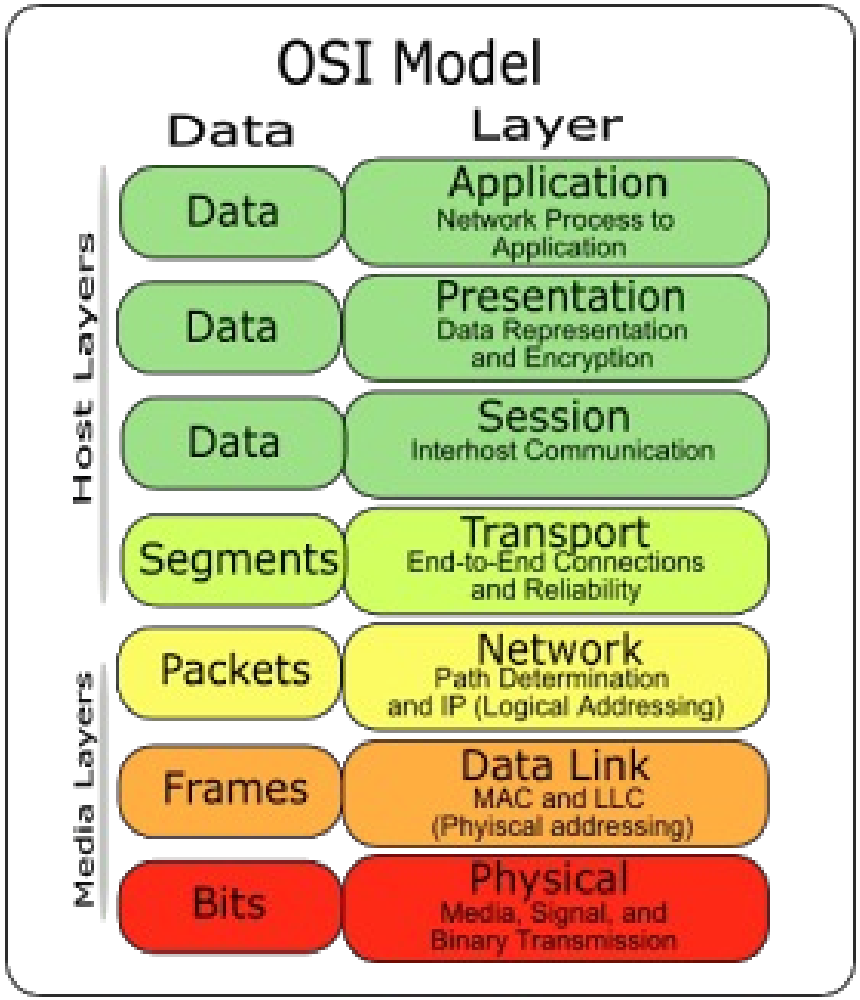
Chapter 7: Networking and IoT

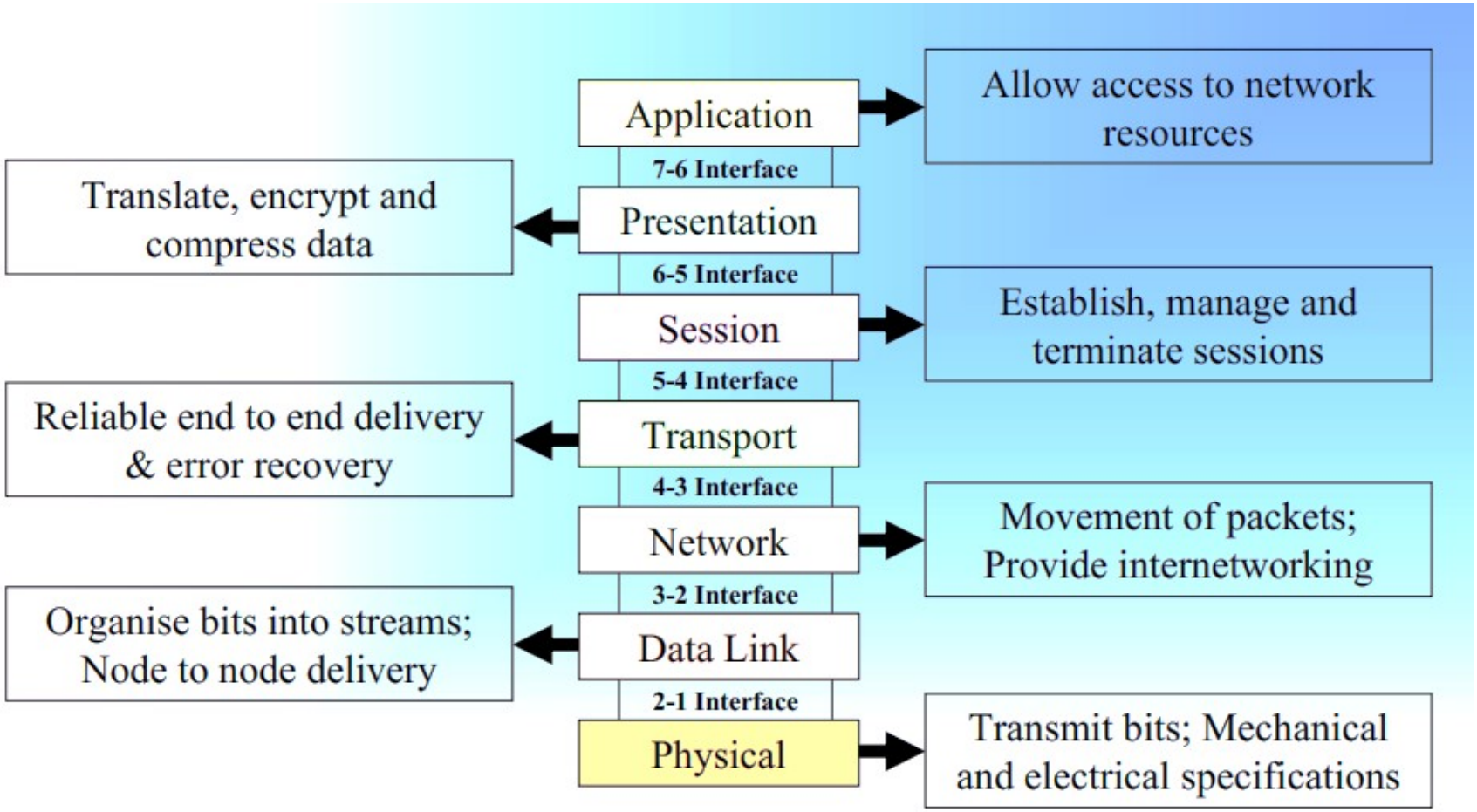
- **Networking and Internet**
  - OSI and TCP/IP model
  - Introduction to communication networks
  - Network devices
  - WWW and HTTP
- **Internet of Things**
  - Introduction to IoT
  - RF Technologies
  - Hardware Platforms
  - Edge Computing
  - Applications

OSI Model

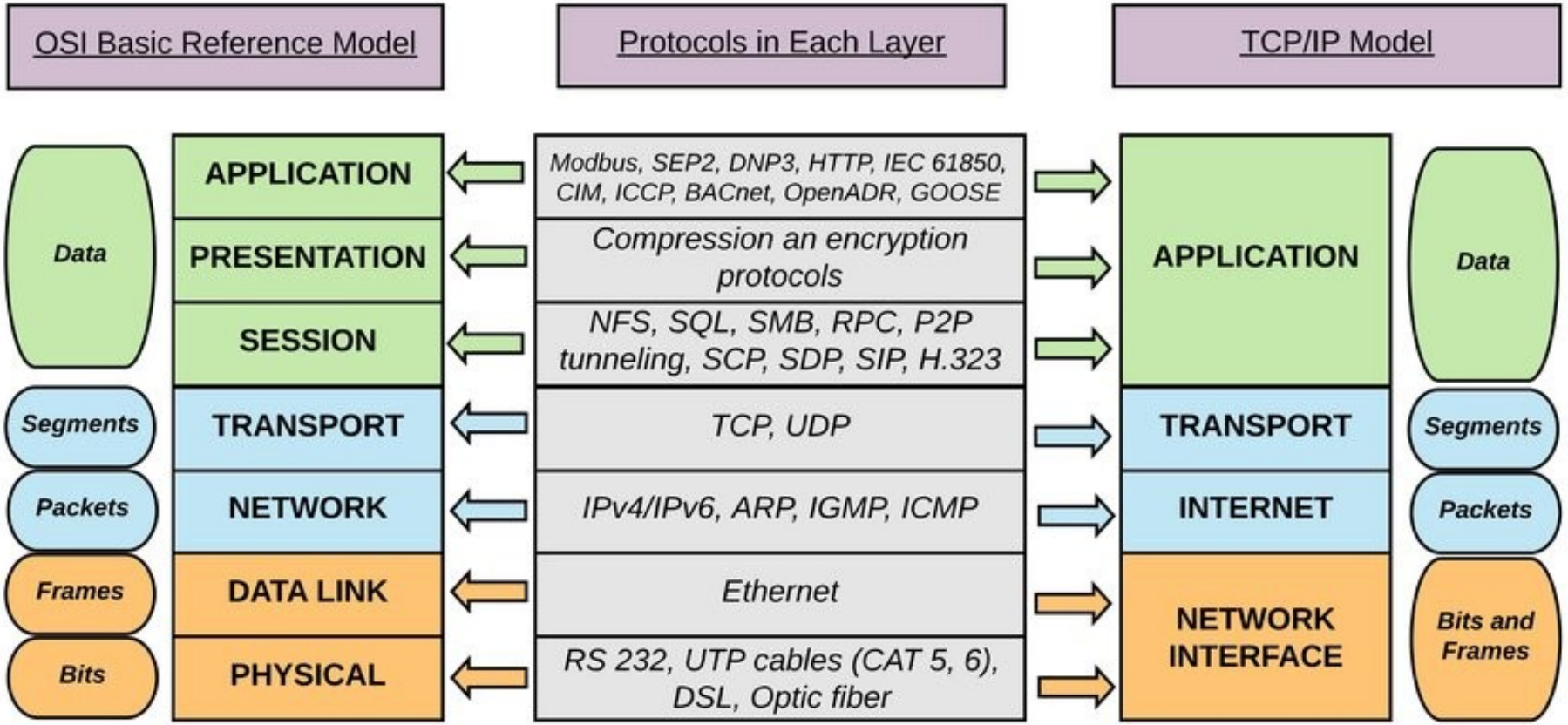
- What is OSI model?
  - Open System Interconnection
- Why do we have to use OSI model?
  - System and Device Design
  - Trouble-shooting
- Layers of OSI model:
  - PHY layer
  - MAC layer
  - Network layer
  - Transport
  - Session layer
  - Presentation layer
  - Application layer



OSI Model: Layer Functions



TCP/IP Model


OSI Basic Reference Model
Protocols in Each Layer
TCP/IP Model
Data
APPLICATION
Modbus, SEP2, DNP3, HTTP, IEC 61850, CIM, ICCP, BACnet, OpenADR, GOOSE
PRESENTATION
Compression an encryption protocols
SESSION
NFS, SQL, SMB, RPC, P2P tunneling, SCP, SDP, SIP, H.323
APPLICATION
Data
Segments
TRANSPORT
TCP, UDP
TRANSPORT
Segments
Packets
NETWORK
IPv4/IPv6, ARP, IGMP, ICMP
INTERNET
Packets
Frames
DATA LINK
Ethernet
NETWORK INTERFACE
Bits and Frames
Bits
PHYSICAL
RS 232, UTP cables (CAT 5, 6), DSL, Optic fiber

Communication Channel

- Communication channels carry data from one computer to another.
- Physical Connections
  - **Twisted-pair** cable consists of pairs of copper wire that are twisted together. Both standard telephone lines and Ethernet cables use twisted pair



  - **Coaxial** cable, a high-frequency transmission cable, replaces the multiple wires of telephone lines with a single solid-copper core



  - **Fiber-optic** cable transmits data as pulses of light through tiny tubes of glass



Communication Channel

- **Wireless Connections**: Wireless connections do not use a solid substance to connect devices. Most use radio waves
  - **Bluetooth**—transmits data over short distances; widely used for a variety of wireless devices.
  - Wi-Fi (wireless fidelity)—uses high-frequency radio signals; most home and business wireless networks use Wi-Fi.
  - **Microwave**—line-of-sight communication; used to send data between buildings; longer distances require microwave stations.
  - **WiMax** (Worldwide Interoperability for Microwave Access)—extends the range of Wi-Fi networks using microwave connections.
  - **Cellular**—uses cell towers to send and receive data within relatively small geographic regions or cells.
  - **Satellite**—uses microwave relay stations; GPS (global positioning system) tracks geographic locations.
  - **Infrared**—uses light waves over a short distance; lineof-sight communication

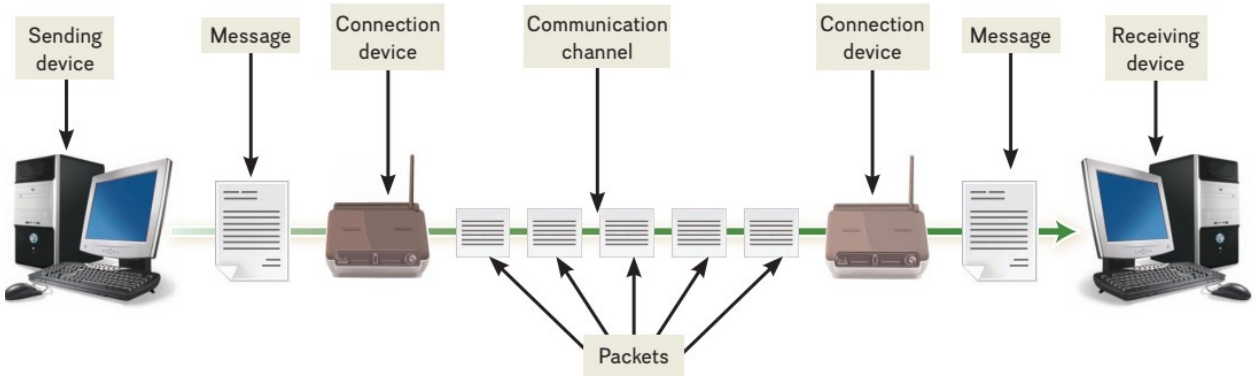


Data Transmission

- **Bandwidth** measures a communication channel's width or capacity. Four bandwidths are *voiceband* (low bandwidth), *medium band*, *broadband* (high-capacity transmissions), and *baseband*.
- **Protocols** are rules for exchanging data. Widely used Internet protocols include https and TCP/IP.
- **IP addresses** (Internet protocol addresses) are unique numeric Internet addresses. DNS (domain name server) converts text-based addresses to and from numeric IPaddresses.
- **Packets** are small parts of messages

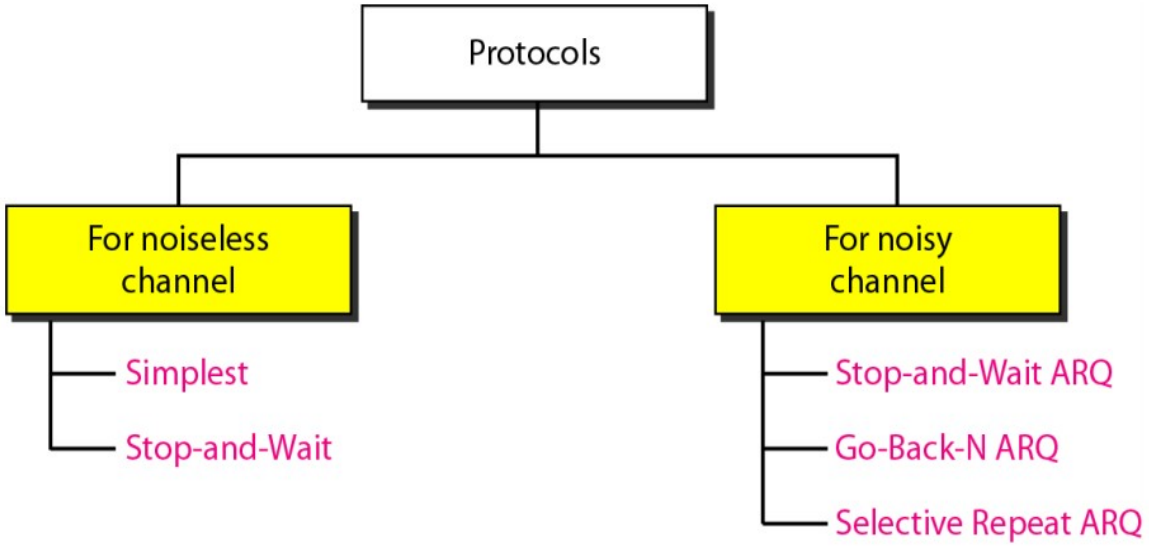

Analog

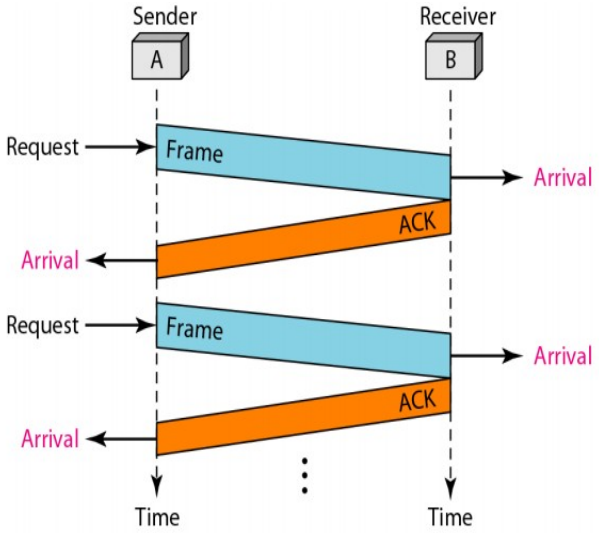

Digital



Communication Protocol

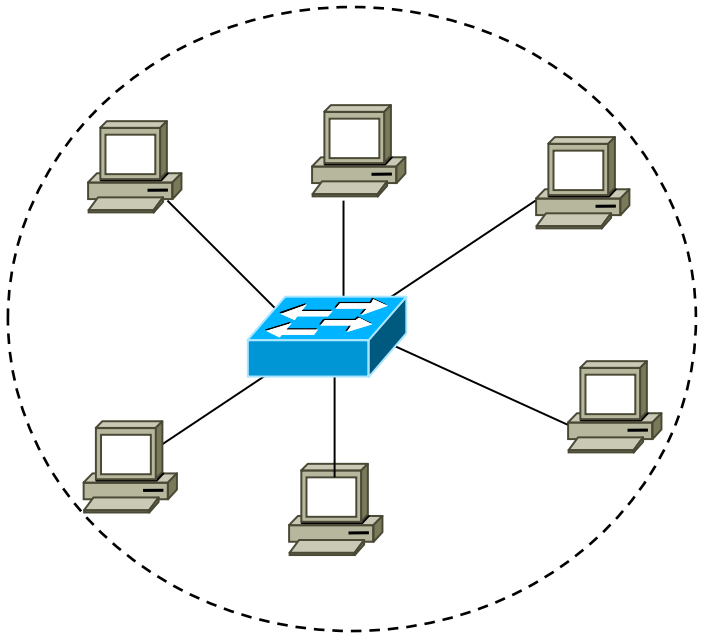
- The combination of **framing**, **flow control**, and **error control** to achieve the delivery of data from one node to another.
- The protocols are normally **implemented in software** by using one of the common programming languages.





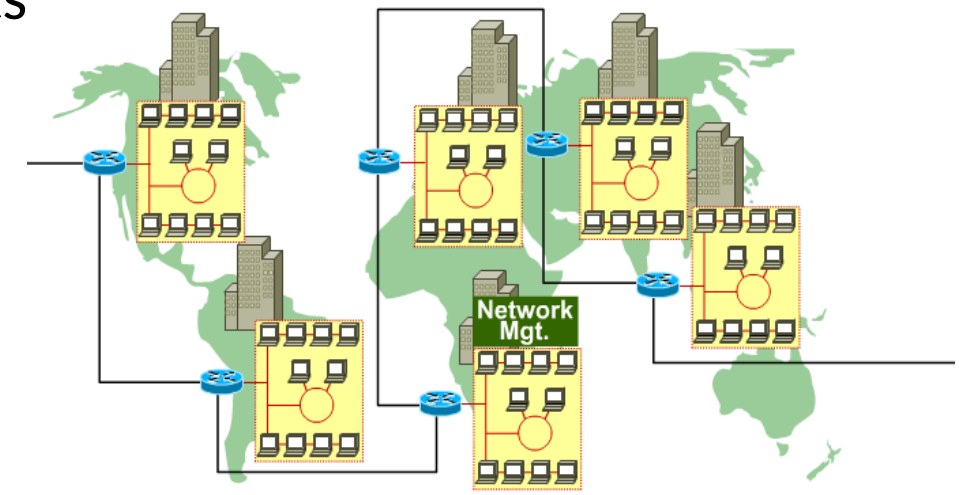
Local Area Network

- A local area network (LAN) is a computer network that interconnects computers in a limited area such as a home, school, computer laboratory, or office building using network media
- Medium:
  - Coaxial cable
  - Twisted-pair line



Wide Area Network

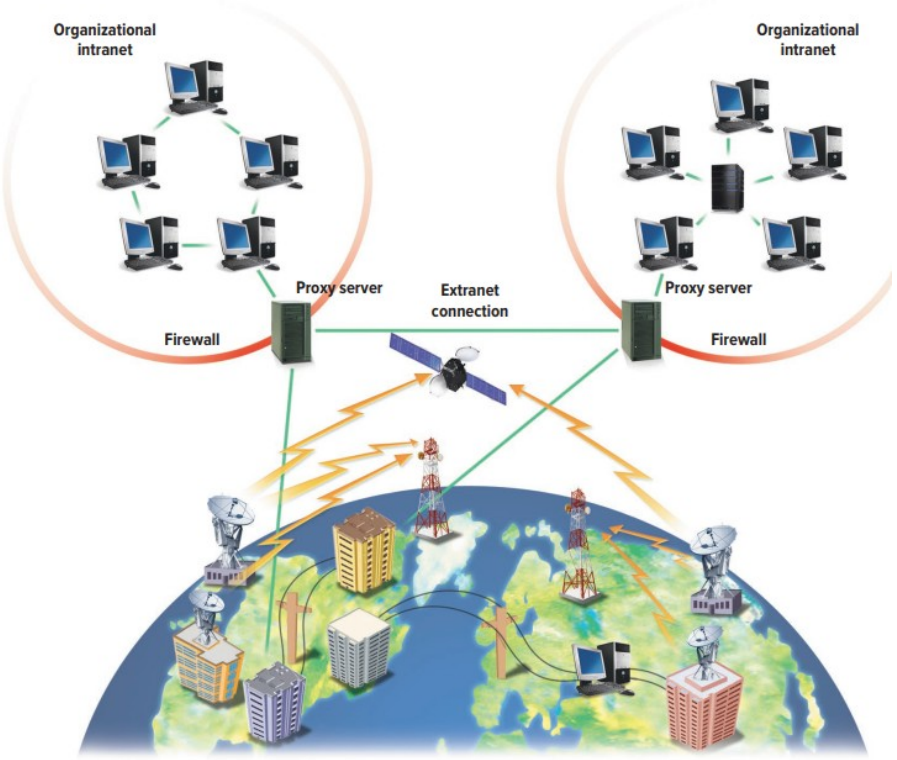
- A Wide Area Network (WAN) is a network that covers a broad area (i.e., any telecommunications network that links across metropolitan, regional, or national boundaries) using private or public network transports
- Besides, several network types:
  - Home
  - WLAN
  - PAN
  - MAN



Type	Description
LAN	Local area network; located within close proximity
Home	Local area network for home and apartment use; typically wireless
WLAN	Wireless local area network; all communication passes through access point
PAN	Personal area network; connects digital devices, such as PDAs
MAN	Metropolitan area network; typically spans cities with coverage up to 100 miles
WAN	Wide area network for countrywide or worldwide coverage

Internet technologies

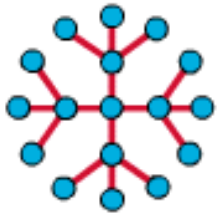
- **Internet** technologies support effective communication using intranets and extranets
  - **Intranet**—*private network* within an organization; uses browsers, websites, and web pages. Typical applications include electronic telephone directories, e-mail addresses, employee benefit information, internal job openings, and much more.
  - **Extranet**—like intranet except connects more than one organization; typically allows suppliers and others limited access to their networks



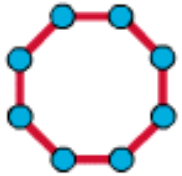
Physical Network Topology



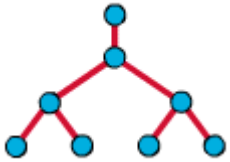
Bus



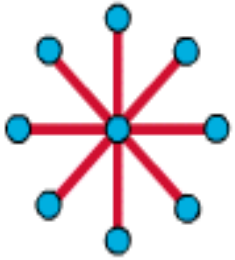
Extended Star



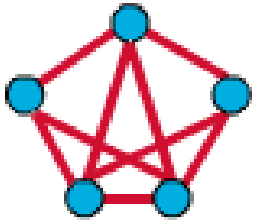
Ring



Hierarchical



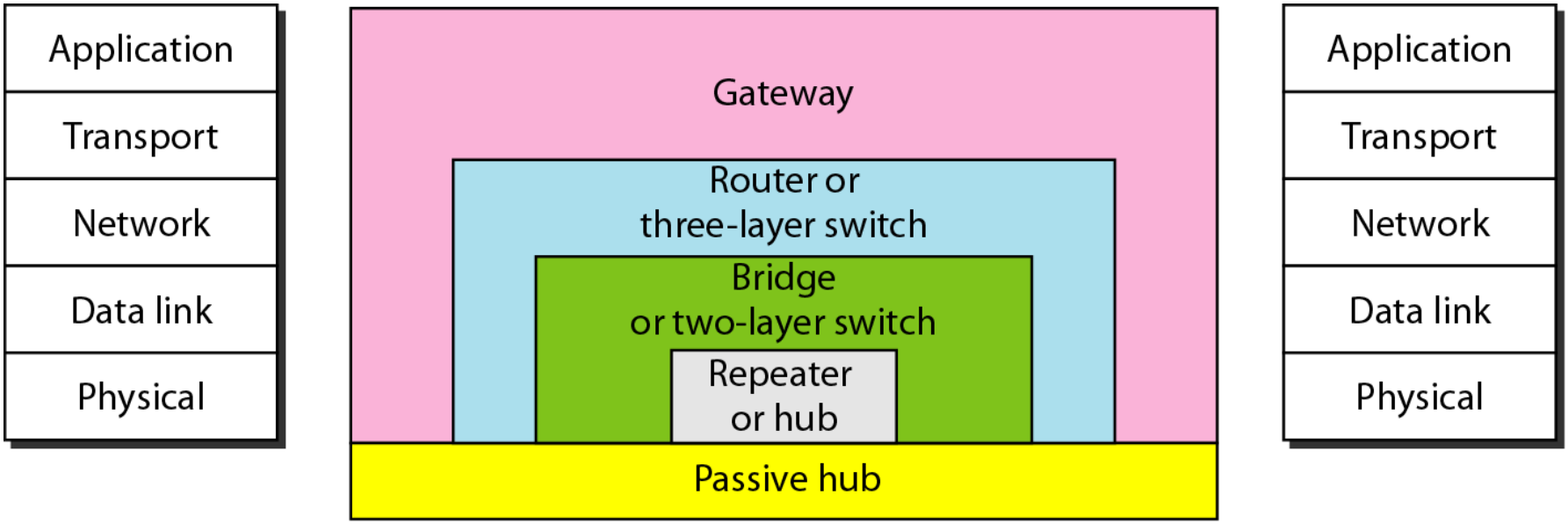
Star



Mesh

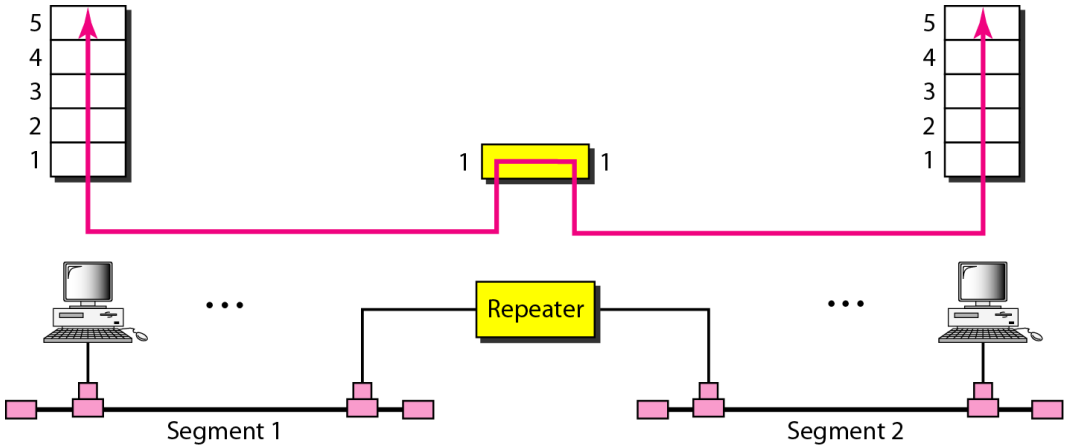
Network Devices

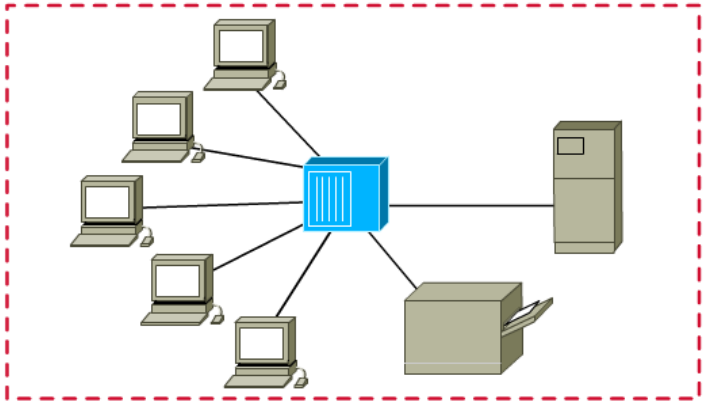
- Hub/Repeater
- Switch/Bridge
- Router/3-layer switch
- Gateway
- Firewall



Network Devices

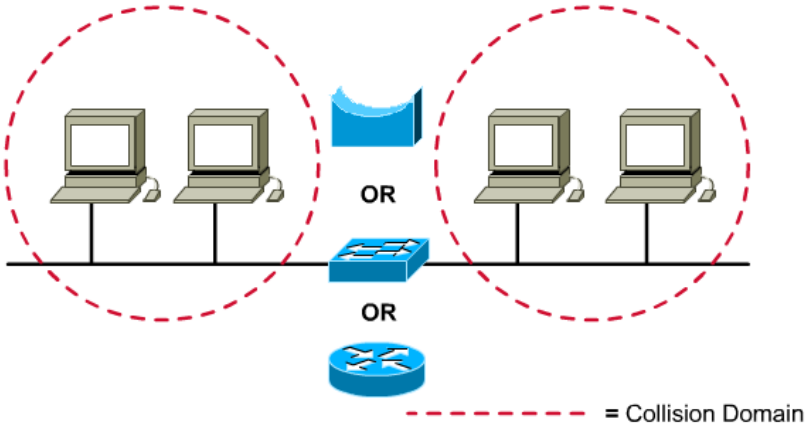
- **Hub/Repeater**
  - Propagate signals from one port to others
  - Layer 1 devices: connects segments of a LAN.
  - Extending a network with a repeater or a hub, results in a larger collision domain.
- Collision Domains are the area where collisions occur. All of layer 1 interconnections are part of the collision domain.

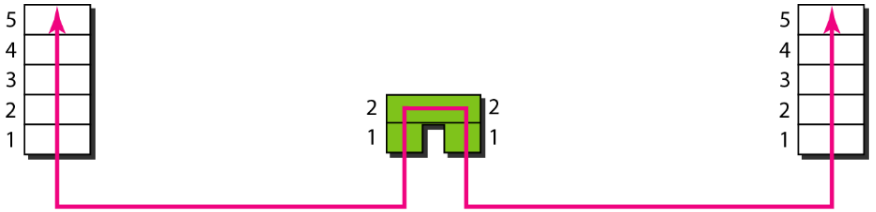




Network Devices

- **Switch**
  - A multi port bridge
  - Support full duplex
  - Hardware switching
  - Dynamically builds and maintains a Content-Addressable Memory (CAM) table
- Segmenting Collision Domain
- Can not prevent layer 2 broadcast domain



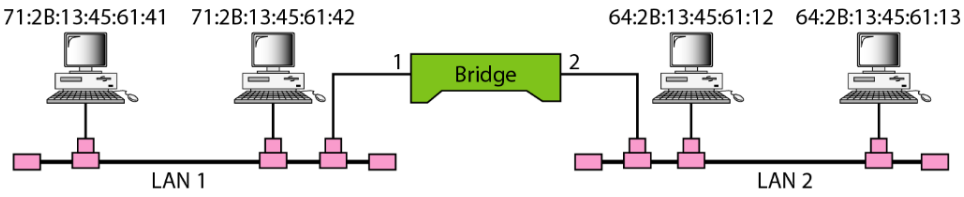


Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	1
64:2B:13:45:61:12	2
64:2B:13:45:61:13	2

Bridge Table



Ethernet Switch



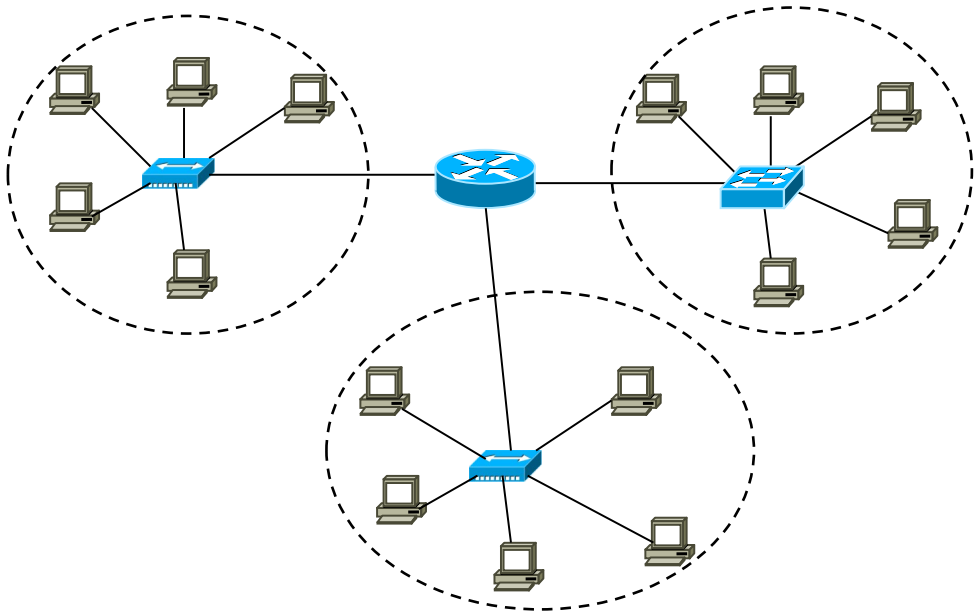
Network Devices

- **Router**:
  - Layer 3 device
  - Routing packets to destination based on destination IP address
  - Running routing protocol
  - Split layer 2 broadcast domain





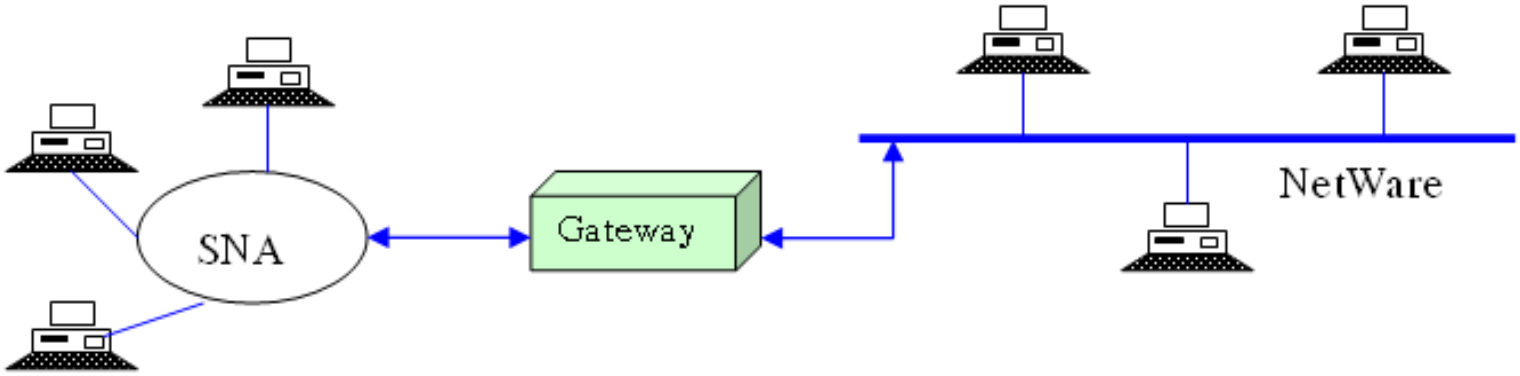
Cisco Router



Network Devices

Gateway:

- The gateway acts as a "Translator" in between two different systems that is used dissimilar communication protocols, data format or different architectures
- Operate at Application layer in OSI model





Synway E1/T1 Trunk Gateway for VoIP

Network Security

- Three technologies commonly used to ensure network security are firewalls, intrusion detection systems, and virtual private networks.
  - **Firewall**—controls access; all communications pass through proxy server.
  - **Intrusion detection systems** (IDS)—work with firewalls; use sophisticated statistical techniques to recognize and disable network attacks.
  - **Virtual private network** (VPN)—creates secure private connection between remote user and organization's internal network



Application architectures

- Client-server
- Peer-to-peer (P2P)
- Hybrid of client-server and P2P

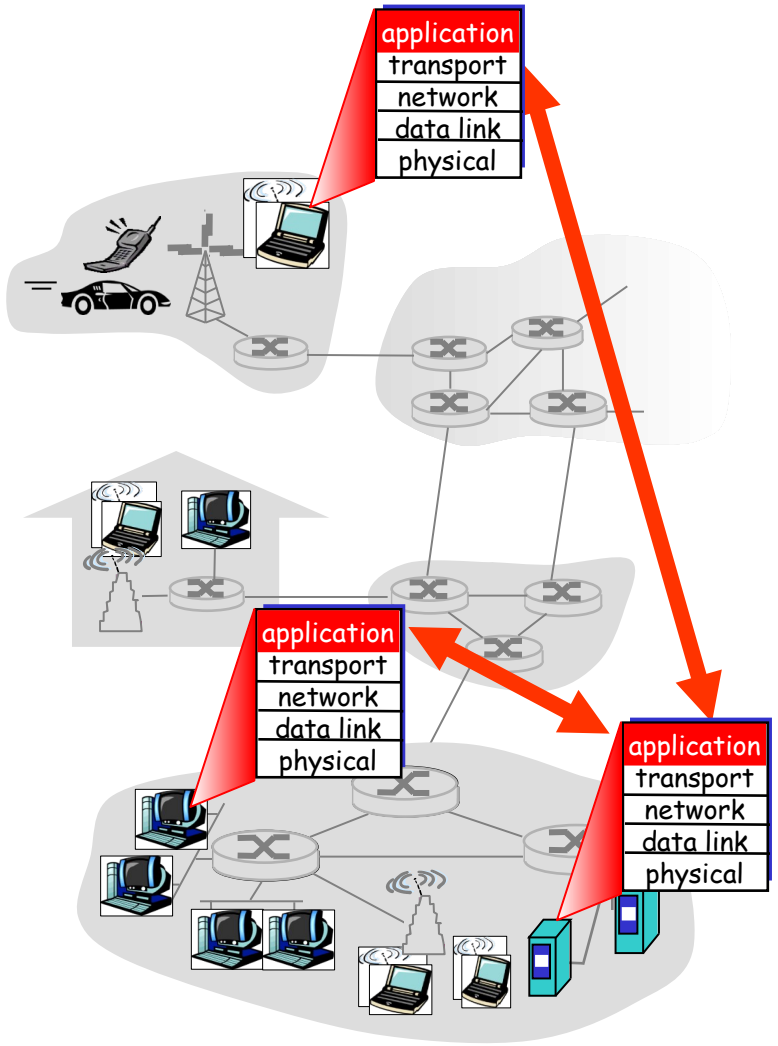
Creating a network app

write programs that

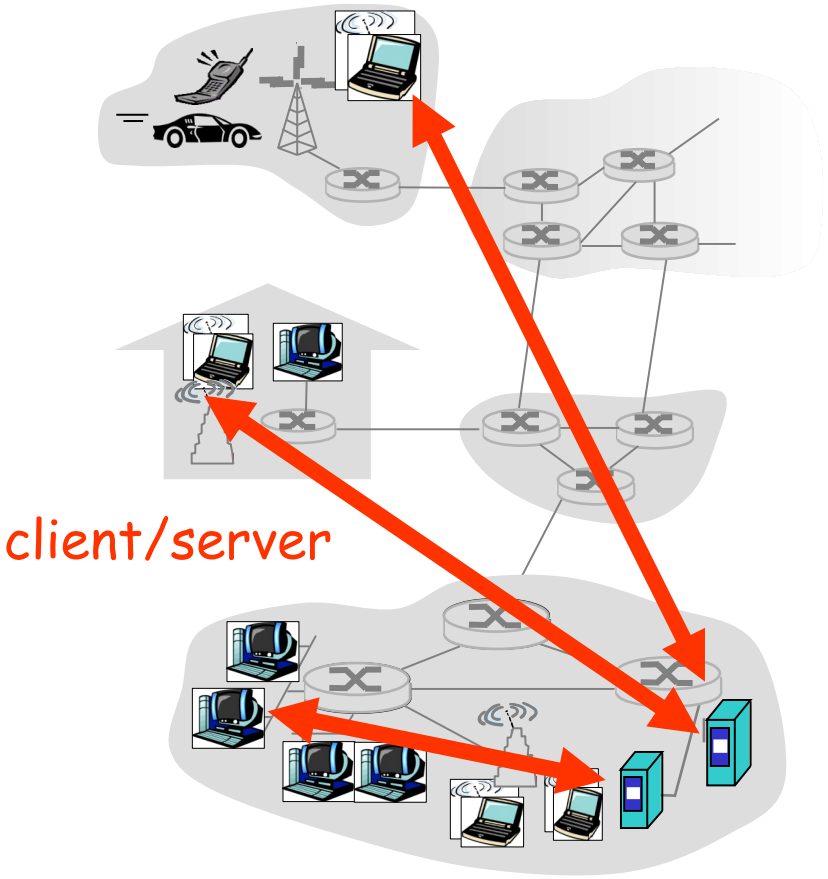
- run on (different) *end systems*
- communicate over network
- e.g., web server software communicates with browser software

No need to write software for network-core devices

- Network-core devices do not run user applications
- applications on end systems allows for rapid app development, propagation



Client-server architecture



server:

- always-on host
- permanent IP address
- server farms for scaling

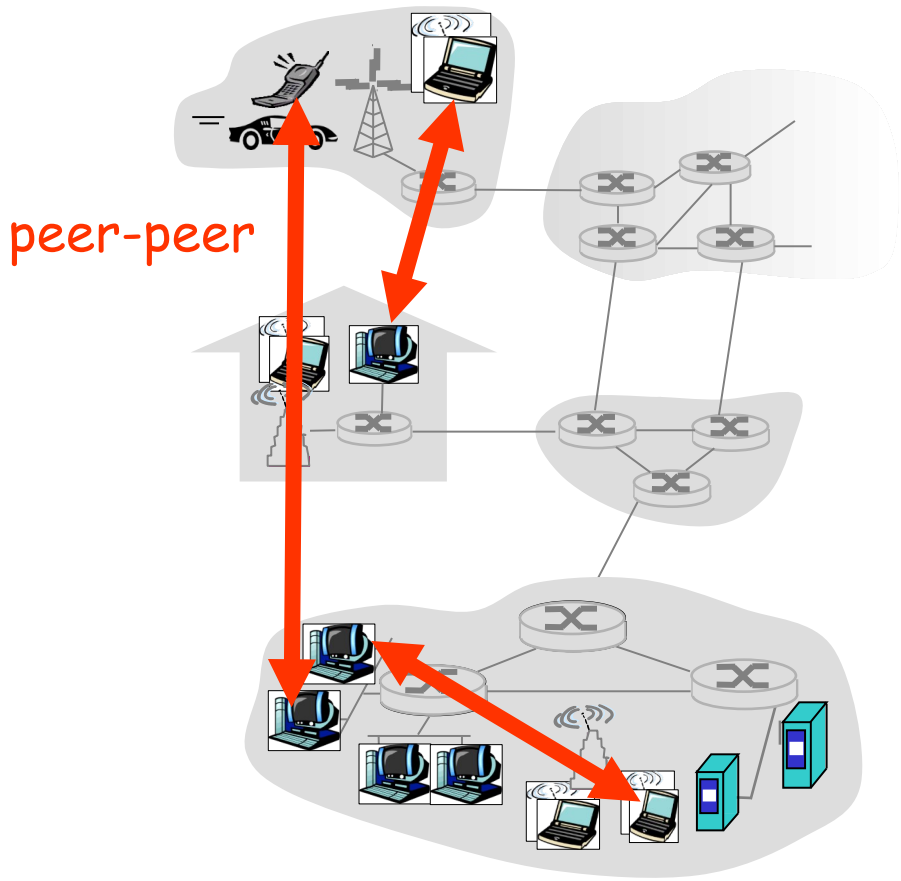
clients:

- communicate with server
- may be intermittently connected
- may have dynamic IP addresses
- do not communicate directly with each other

Pure P2P architecture

- *no* always-on server
- arbitrary end systems directly communicate
- peers are intermittently connected and change IP addresses

Highly scalable but difficult to manage



Processes communicating

Process: program running within a host.

- within same host, two processes communicate using inter-process communication (defined by OS).
- processes in different hosts communicate by exchanging messages

Client process: process that initiates communication

Server process: process that waits to be contacted

r Note: applications with P2P architectures have client processes & server processes

App-layer protocol defines

- Types of messages exchanged,
  - e.g., request, response
- Message syntax:
  - what fields in messages & how fields are delineated
- Message semantics
  - meaning of information in fields
- Rules for when and how processes send & respond to messages

Public-domain protocols:

- defined in RFCs
- allows for interoperability
- e.g., HTTP, SMTP

Proprietary protocols:

- e.g., Skype

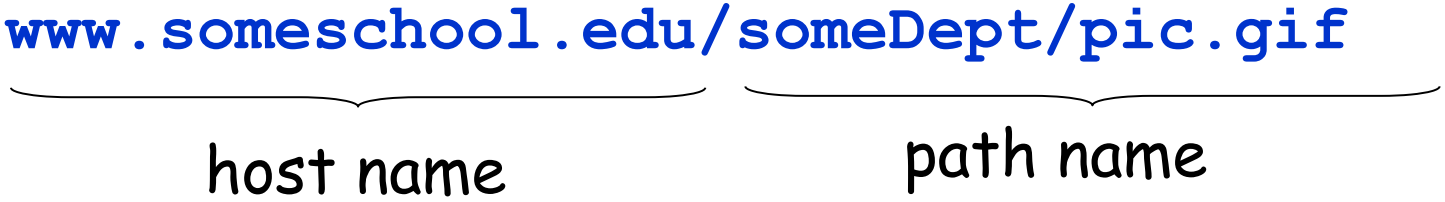
Internet apps: application, transport protocols

Application	Application layer protocol	Underlying transport protocol
e-mail	SMTP [RFC 2821]	TCP
remote terminal	Telnet [RFC 854]	TCP
access	HTTP [RFC 2616]	TCP
Web	FTP [RFC 959]	TCP
file transfer	HTTP (eg Youtube),	TCP or UDP
streaming multimedia	RTP [RFC 1889]	
Internet telephony	SIP, RTP, proprietary (e.g., Skype)	typically UDP

Web and HTTP

First some jargon

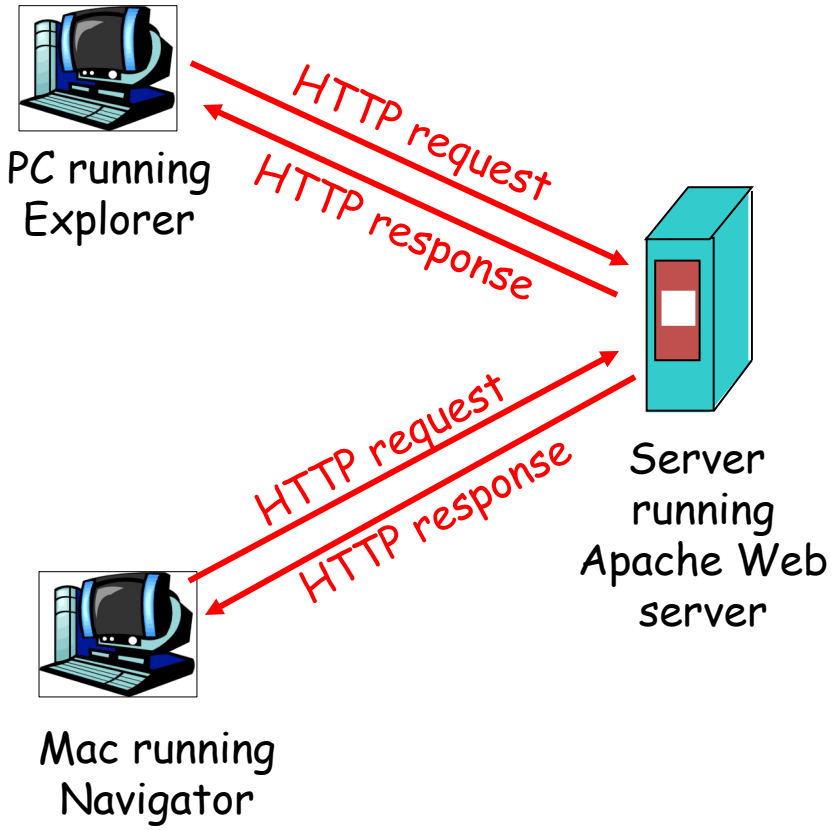
- Web page consists of objects
- Object can be HTML file, JPEG image, Java applet, audio file,…
- Web page consists of base HTML-file which includes several referenced objects
- Each object is addressable by a URL
- Example URL:



HTTP overview

HTTP: hypertext transfer protocol

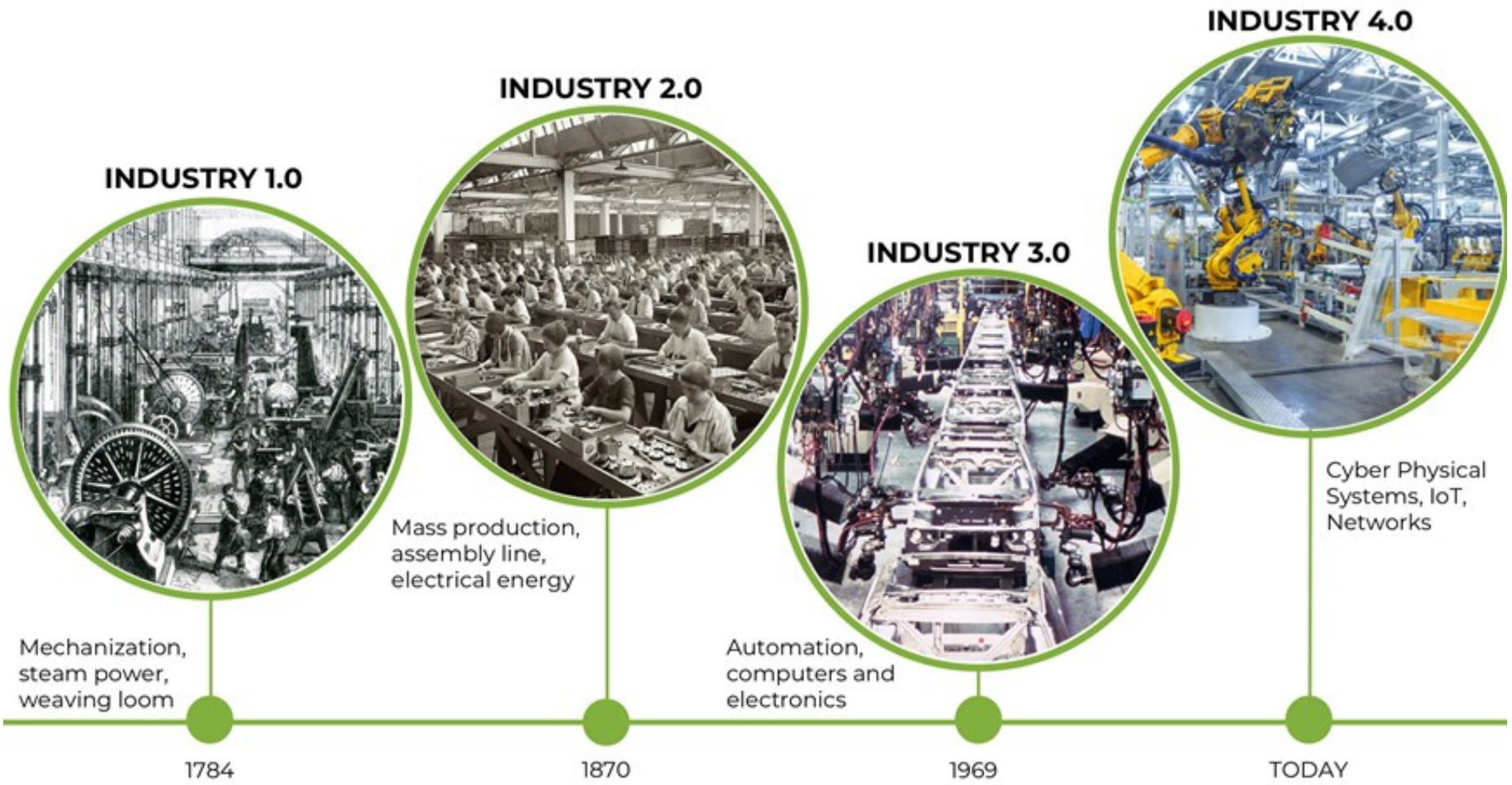
- Web's application layer protocol
- client/server model
  - *client:* browser that requests, receives, "displays" Web objects
  - *server:* Web server sends objects in response to requests



Internet of Things

- Introduction to IoT
- RF Technologies
- Hardware Platforms
- Edge Computing
- Applications

Industry 4.0 and Internet of Things



Internet of Things (IoT) with AI

(Timothy Chou)

Do

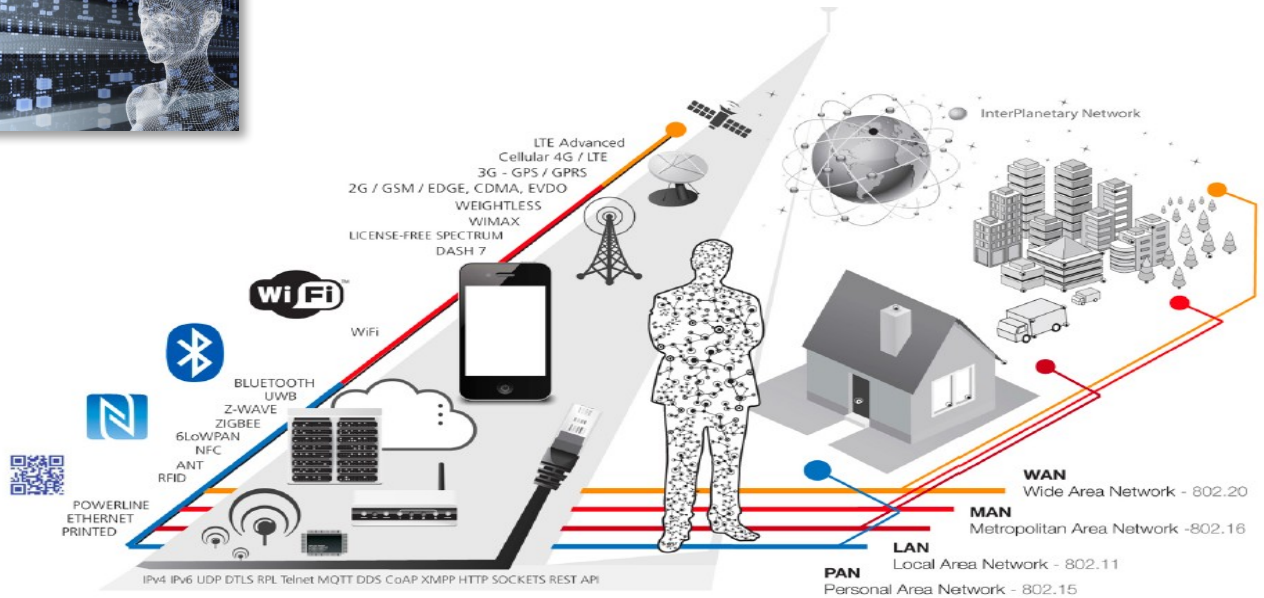




Learn



Collect

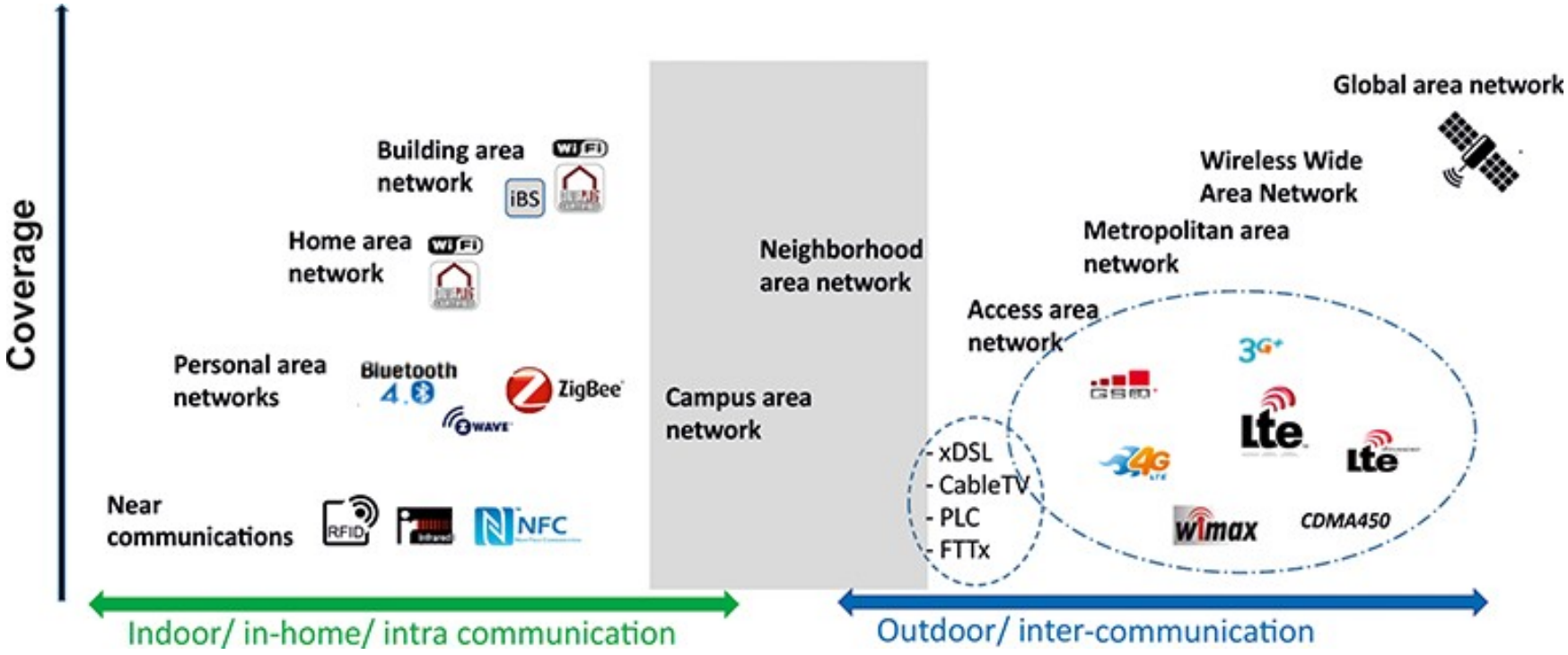


Connect

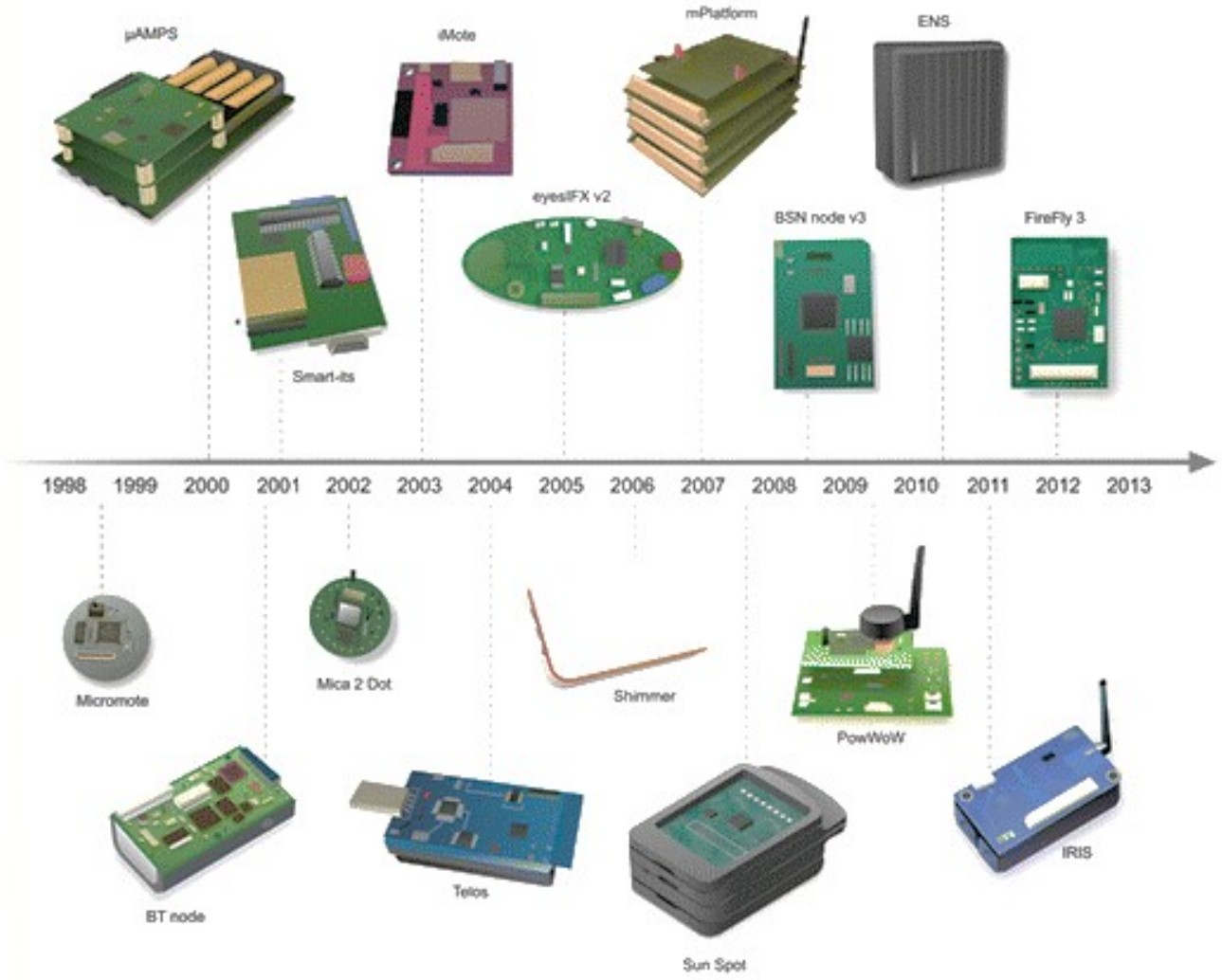
Things



Connect Layer

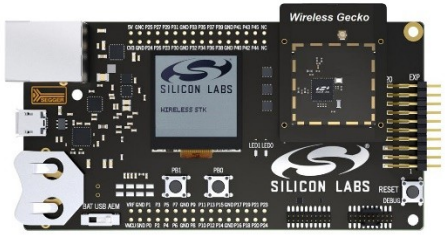


IoT End-devices



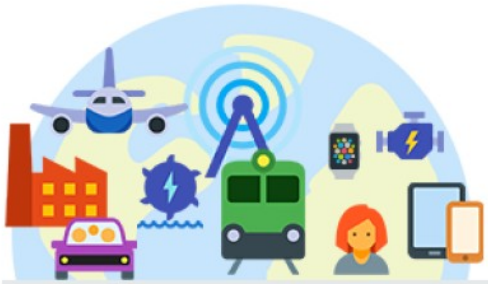








Edge Computing



Operations Technology

Information Technology

THE EDGE

The Things

- Wearables
- Mobile Devices
- Cars
- Meters
- Motors
- Robotics
- Buildings
- Generators

Sensors & Actuators

Data Aggregation & Gateways

Edge IT

Data Center & Cloud IT

DATA GENERATION

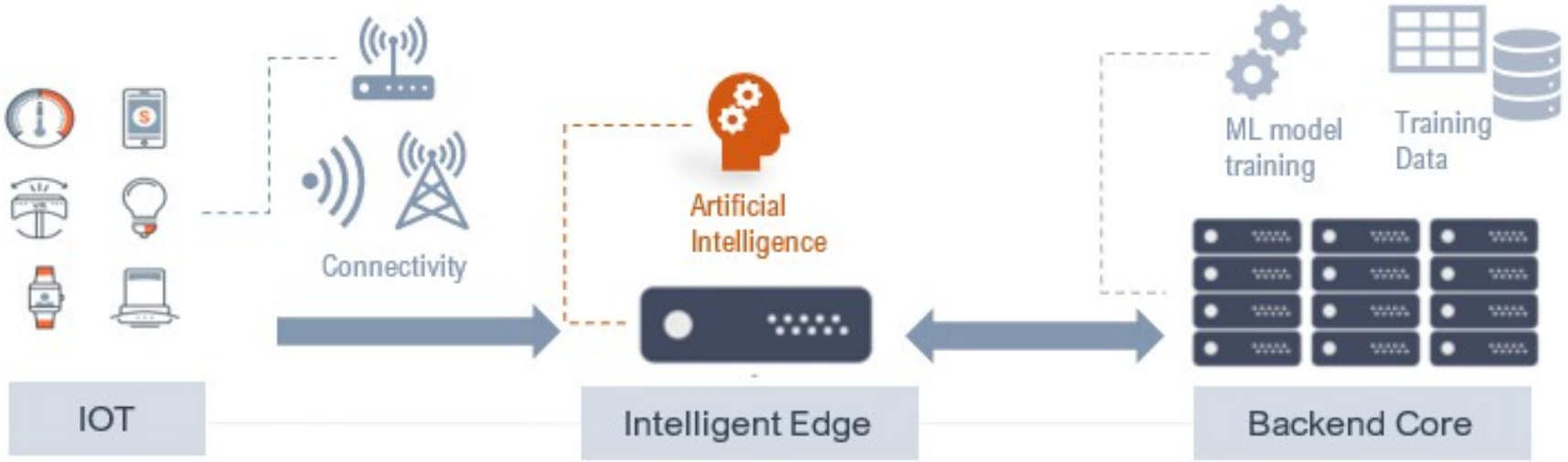
DATA SENSING

DATA COLLECTION AGGREGATION

EARLY DATA ANALYTICS

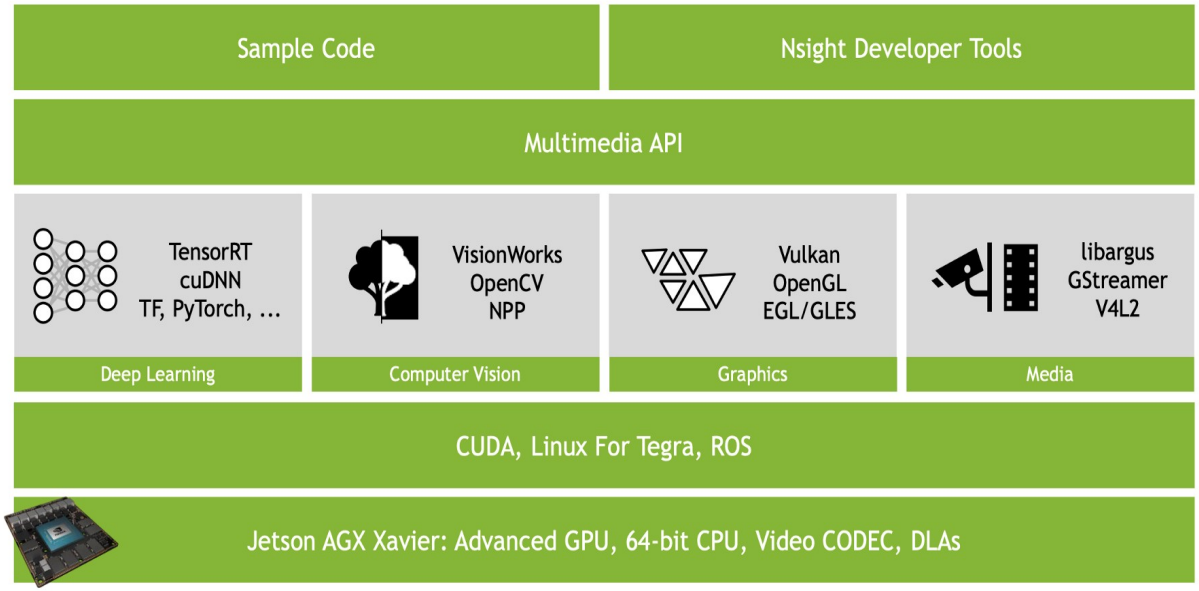
DEEP DATA ANALYTICS

Combine it with AI = Edge AI



- Edge AI overcomes the latency of data and **complex computations:**
  - ❖Increase in levels of Automation
  - ❖Digital Twins for Advanced Analytics
  - ❖Real-Time Decision Making
  - ❖Edge Inference and Training

Edge AI on Embedded Platform



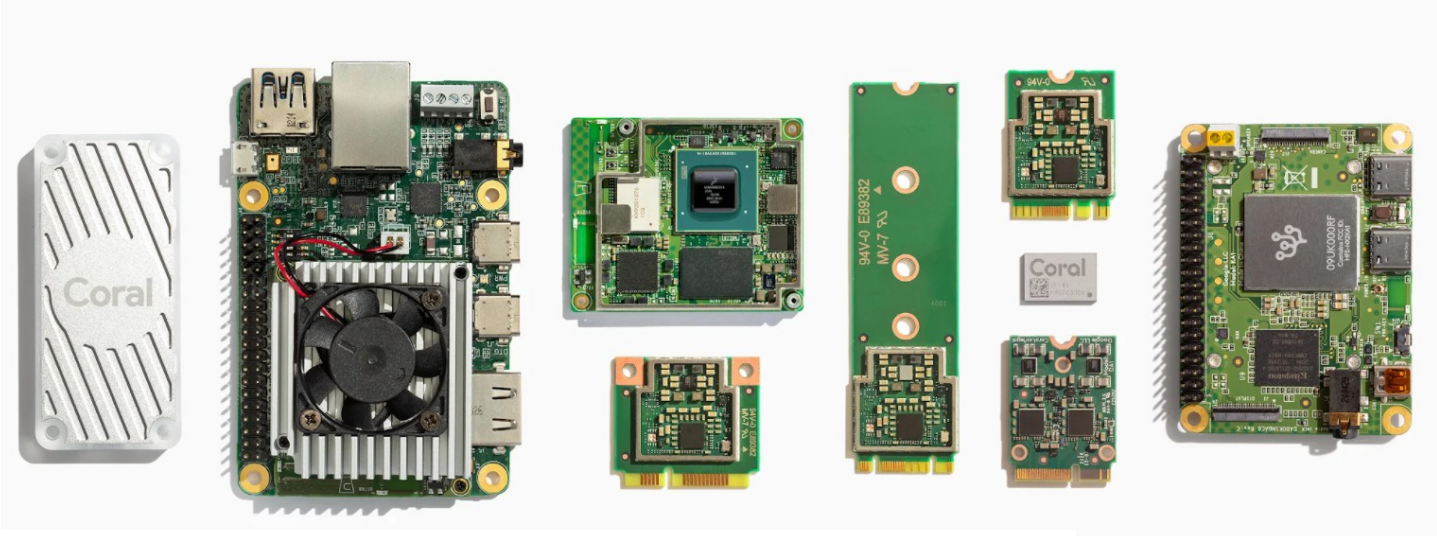


LLM on target device



384 NVIDIA CUDA cores and
48 Tensor Cores

PyCoral [https://coral.ai/products/]



Models

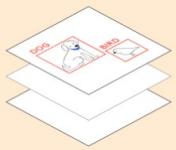
Trained TensorFlow models for the Edge TPU



Image classification

Models that recognize the subject in an image, plus classification models for on-device transfer learning.

→ See models



Object detection

Models that identify multiple objects and provide their location.

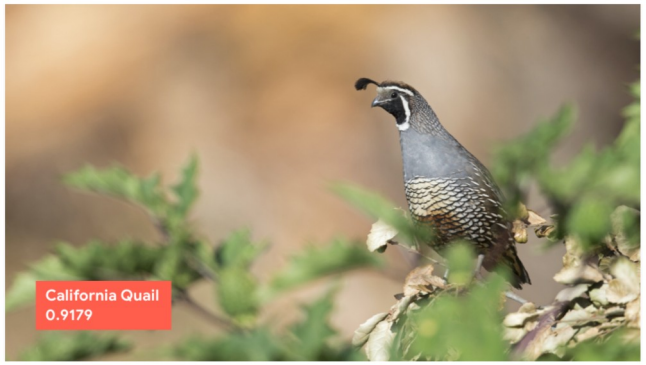
→ See models



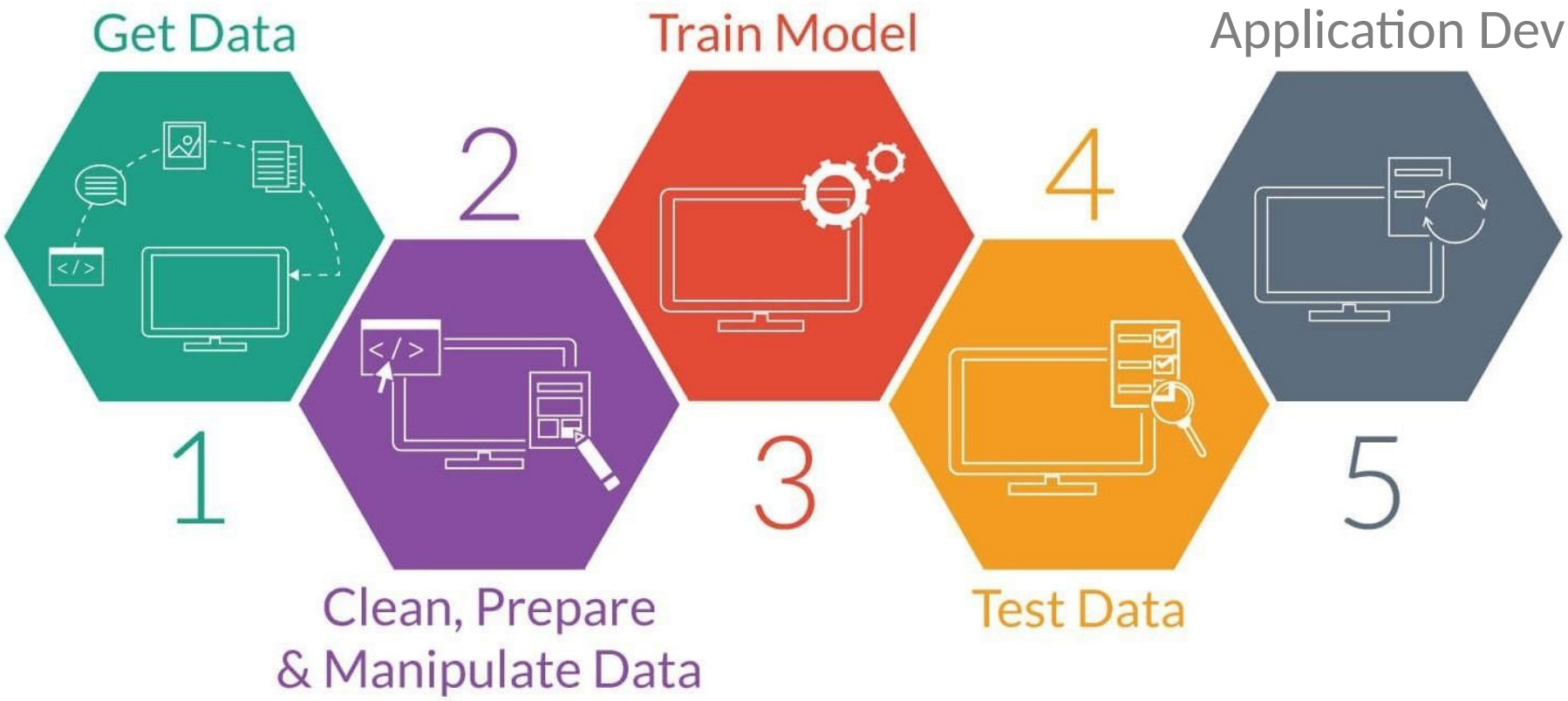
Semantic segmentation

Models that identify specific pixels belonging to different objects.

→ See models

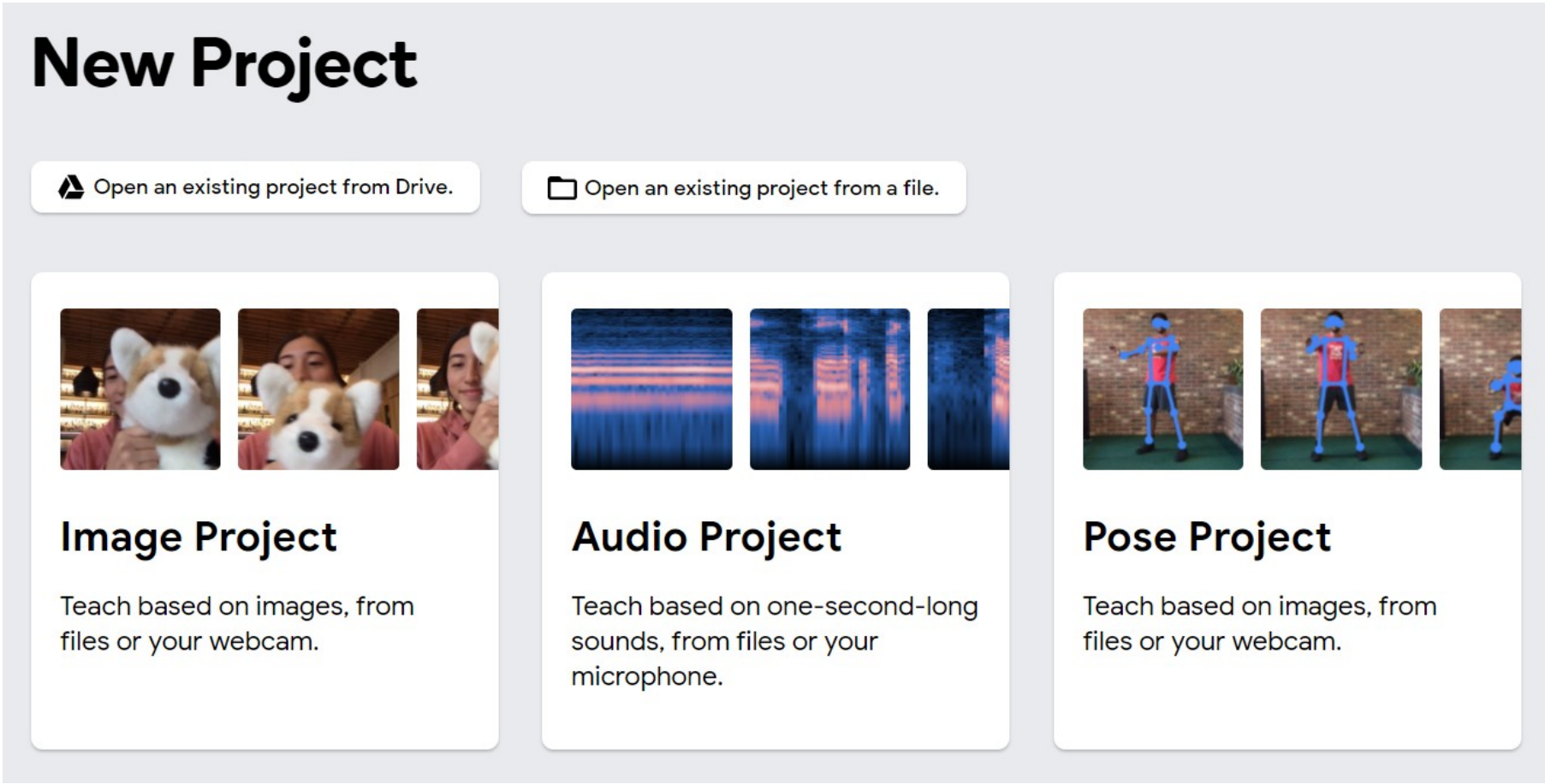


Train an AI Model



Demo with Google TM

- https://teachablemachine.withgoogle.com/train



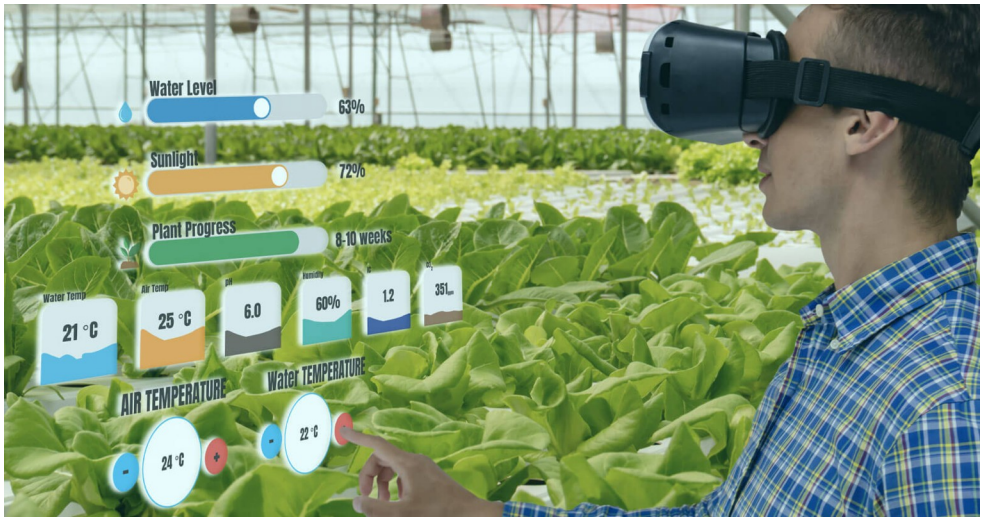
Smart Agriculture

- Temperature and Humidity
- pH, EC
- Light intensity
- Pump controllers





Monitoring system based on AR/VR



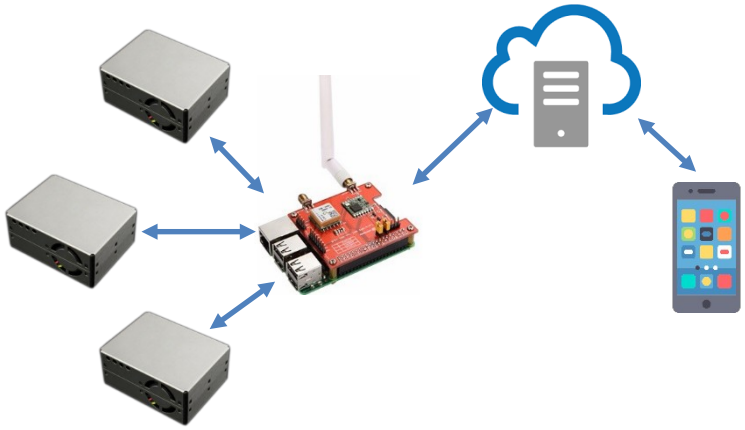






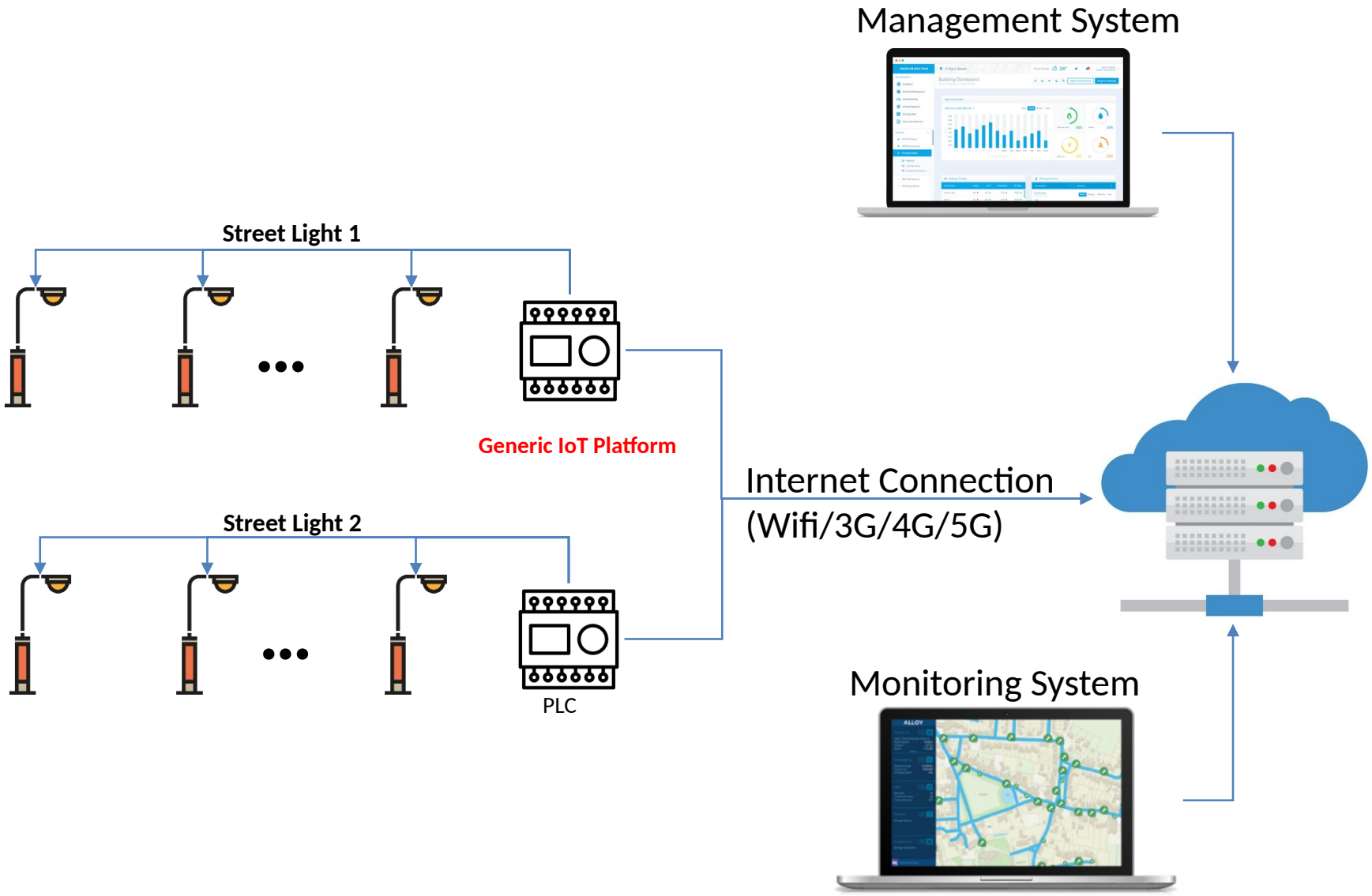
Air Quality Monitoring





- Air quality monitoring program assists us in improving and developing air pollution control programs to reduce the effect of air pollution.
- PM2.5, PM10, CO2, CO

Smart Street Light



Applications: Internet of Things

